



Course Title: Big Data Processing

Course Code: EGCI 466

Undergraduate Program

Mahidol University International College

Division Science

TQF 3 Course Specifications

Section 1 General Information

1. Course code and course title

Thai	EGCI 466	การประมวลผลข้อมูลปริมาณมาก
English.	EGCI. 466	Big Data Processing

2. Number of credits 4 (4-0-8)

3. Program and type of subject

3.1 Program Bachelor of Engineering (Computer Engineering)

3.2 Type of Subject Major Course (Required Major)

4. Course Coordinator and Course Lecturer

4.1 Course Coordinator Lect. Dr. Mingmanas Sivaraksa

4.2 Course Lecturer Lect. Dr. Mingmanas Sivaraksa

5. Trimester/ Year of Study

5.1 Trimester First trimester / for 2nd year Computer Engineering

5.2 Course Capacity Approximately 20 students

6. Pre-requisite EGCI 221 Data Structures and Algorithms

7. Co-requisites None

8. Venue of Study Mahidol University, Salaya campus

9. Date of Latest Revision

Date: 13 July 2020



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Section 2 Goals and Objectives

1. Course Goals

After successful completion of this course, students will be able to:

- Describe the principles of big data processing
- Choose tools that is appropriate for each data types
- Apply the knowledge and able to understand various problems using Big Data processing knowledge

2. Objectives of Course Development/Revision

2.1 Course Objectives

Students should able to understand and use big data processing: map reduce, distribution techniques to solve data problems in other field. They should have enough knowledge to pursue new knowledge, develop, choose tools and evaluate the validity of choice using big data processing knowledge.

2.2 Course-level Learning Outcomes: CLOs

By the end of the course, students will be able to (CLOs)

- CLO1: Use Big Data Theory to solve problems in other fields
- CLO2: Pursue new knowledge from Big data concepts
- CLO3: Choose Big Data tools and techniques for development



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Section 3 Course Management

1. Course Description

(Thai) ทฤษฎี และวิธีการจัดการข้อมูลปริมาณมาก การจัดเก็บข้อมูลปริมาณมาก ระบบแบบกระจายในการจัดการข้อมูลปริมาณมาก การทำเหมืองข้อมูลปริมาณมาก อัลกอริทึมและการติดตั้งใช้งานในระบบแบบกระจาย การแสดงผลข้อมูลปริมาณมาก แนวโน้มและการประยุกต์ใช้ข้อมูลปริมาณมาก

(English) Theory and techniques for managing big data, collecting big data, distribution system for big data, mining big data, algorithms and their implementation on distributed systems; big data visualization; trends and applications for big data

2. Credit hours / trimester

Lecture (hours)	Additional Class (hours)	Laboratory/field trip/internship (hours)	Self-study (hours)
48 hours (4 hours x 12 weeks)	-	-	96 hours (8 hours x 12 weeks)

3. Numbers of hours that the lecturer provides individual counseling and guidance

1 hour/week



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Section 4 Development of Students' Learning Outcome

1. Short summary on the knowledge or skills that the course intends to develop in students (CLOs)

By the end of the course, students will be able to

CLO1: Use Big Data Theory to solve problems in other fields

CLO2: Pursue new knowledge from Big data concepts

CLO3: Choose Big Data tools and techniques for development

2. Teaching methods for developing the knowledge or skills specified in item 1 and evaluation methods of the course learning outcomes

Course Code	Teaching methods	Evaluation Methods
CLO1	Interactive Lecture, Individual Assignment	Written Examination, Individual Evaluation
CLO2	Interactive Lecture, Individual Programming Assignment, Project Assignment	Written Examination, Project Evaluation, Individual Evaluation
CLO3	Interactive Lecture, Case Study, Project Assignment	Written Examination, Project Evaluation, Individual Evaluation



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Section 5 Teaching and Evaluation Plans

1. Teaching plan

Week	Topic	Number of Hours		Teaching Activities/ Media	Evaluation
		Lecture Hours	Lab/Field Trip/Internship Hours		
1	Big data principle and theory	4	0	Interactive lecture and exercises	Interactive Lecture
2	Big data access and storage	4	0	Interactive lecture and exercises	Interactive Lecture, Individual Programming Assessment
3	Mapreduce theory and application	4	0		
4	Mapreduce Application	4	0		
5	Link Analysis	4	0		
6	Similarity analysis and	4	0	Interactive lecture and Group Assignment	Midterm Assessment
7	Midterm Review & Assessment	2	-		Written Exam
7-8	Data Stream Mining	4	0	Lecture and homework	Interactive Lecture, Individual Programming Assessment
8-9	Analysis of Large Graphs	6	0	Lecture and homework	Interactive Lecture, Individual



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					Assessment
10-11	Decision Tree	4	0	Lecture and homework	Interactive Lecture, Individual Programming Assessment
11	Page Rank	4	0	Lecture and homework	Final Assessment
12	Current trends in big data	4	0	Discussion and presentation	Interactive Lecture, Project Assessment
	Total	48	0		

2. Plan for Assessing Course Learning Outcomes

2.1 Assessing and Evaluating Learning Achievement

a. Formative Assessment

The assessment tools such as exercises and assignments are used to evaluate student's understanding by their ability to choose, create and evaluate pattern recognition tools and techniques. Group assignment aims to build skills in pursuing new knowledges.

b. Summative Assessment

(1) Tools and Percentage Weight in Assessment and Evaluation

Learning Outcomes	Assessment Methods	Assessment Ratio (Percentage)	
CLO1: Use Big Data Theory to solve problems in other fields	Individual Assignment	15	35
	Midterm Examination	10	
	Final Examination	10	



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CLO2: Pursue new knowledge from Big data concepts	Individual Assignment	15	35
	Midterm Examination	10	
	Group Project	10	
CLO3: Choose Big Data tools and techniques for development	Individual Assignment	10	35
	Midterm Examination	5	
	Final Examination	10	
	Group Project	10	
Total			100

(2) Grading System

Grade	Achievement	Final Score (% range)	GPA
A	Excellent	90-100	4.0
B+	Very Good	85-89	3.5
B	Good	80-84	3.0
C+	Fairly Good	75-79	2.5
C	Fair	70-74	2.0
D+	Poor	65-69	1.5
D	Very Poor	60-64	1.0
F	Fail	Less than 60	0.0

(3) Re-examination (If course lecturer allows to have re-examination)

N/A - (Not applicable with MUIC)

3. Student Appeals

The student wishing to appeal according to grading result must submit a written and signed appeal form personally to the academic affair unit. It is prohibited to assign another person to appeal on one's behalf. The written appeal form is then sent to the program director



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and chair of department. The final decision is transferred for approval by the faculty committee. The result of appeal then is informed to the student.



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Section 6 Teaching Materials and Resources

1. Textbooks and/or other documents/materials

- White, T. (2014). Hadoop: The definitive guide. " O'Reilly Media, Inc."
- Tanenbaum, A., & Van Steen, M. (2007). Distributed systems. Pearson Prentice Hall.
- Rajaraman, A., & Ullman, J. D. (2011). Mining of massive datasets. Cambridge University Press.

2. Recommended textbooks and/or other documents/materials

3. Other Resources (If any)

<http://www-03.ibm.com/software/products/en/category/bigdata>



Section 7 Evaluation and Improvement of Course Management

1. Strategies for evaluating course effectiveness by students
 - 1.1 Evaluation of peers by students
 - 1.2 Student evaluation
 - 1.2.1 Course content
 - 1.2.2 Course management
 - 1.2.3 Suggestions
 - 1.2.4 Overall opinions
2. Strategies for evaluating teaching methods
 - 2.1 Student evaluation
 - 2.2 Presentation
3. Improvement of teaching methods

Use evaluation from 1 and 2 for course improvement.
4. Verification process for evaluating students' standard achievement outcomes in the course

Analysis of students' learning outcomes using scores from each CLOs for evaluation.
5. Review and plan for improving the effectiveness of the course

Review the course before trimester starts, before each teaching period and review course contents every 3 years.



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Table 1 The relationship between course and Program Learning Outcomes (PLOs)

	Program Learning Outcomes (PLOs)					
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6
EGCI 466 Big data Processing		R	R			R

Note: Indicate the level of CLOs by letter I, R, P or M. Using the information as shown in the Curriculum Mapping of TQF2

Table 2 The relationship between CLOs and PLOs

EGCI 463	Program Learning Outcomes (PLOs)					
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6
CLO1: Use Big Data Theory to solve problems in other fields		2.1				
CLO2: Pursue new knowledge from Big data concepts		2.2				
CLO3: Choose Big Data tools and techniques for development						6.1

Table 3 The description of PLOs and Sub Los of the course

PLOs	SubPLOs
PLO2. Integrate computer engineering knowledge with other related sciences and pursue new knowledge in computer engineering.	2.1 Use computer engineering knowledge to solve problems in other fields
	2.2 Peruse new knowledge in computer engineering using other related science
PLO6. Create a related computer engineering development based on information technologies in mathematics or applied statistics.	6.1 Choose information technology tools properly for computer engineering development.